

1 **Null Simulations Quantify False Catamenial Epilepsy Classification**

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17

Abstract

Objective: To quantify how often standard and historical catamenial epilepsy classification rules produce apparent catamenial epilepsy under a null model in which realistic seizure diaries and realistic menstrual-cycle diaries are generated independently. Methods: We simulated 100,000 synthetic participants for 36 months, split a priori into healthy ovulatory and broad population cohorts. CHOCOLATES generated seizure diaries and HORMONE-CYCLE generated independent menstrual/hormone diaries. A random circular shift was applied before merging. Herzog phases were labeled before window sampling. We evaluated exact Herzog 2004, windowed Herzog thresholds, minimum-data rules, cycle reproducibility rules, negative-binomial regression, and assumption-based historical definitions across prespecified calendar, cycle, and full-diary windows. Results: In 36-month windows, windowed Herzog thresholds classified 11.2% (95% CI 11.0-11.5) of healthy ovulatory and 36.3% (35.8-36.7) of broad population participants as false-positive catamenial epilepsy. Negative-binomial regression reduced false positives to 4.1% (3.9-4.2) and 4.2% (4.0-4.4), respectively. Exact Herzog 2004 applied to 3-cycle windows classified 50.3% and 51.6% of classifiable windows positive but left 53.8% and 66.7% indeterminate. In 10,000 null studies of 30 participants using 3-month windows, apparent prevalence exceeded 39.1% in 45.9% of healthy ovulatory and 79.9% of broad population studies. Conclusions: Chance alignment between realistic seizure cyclicality and menstrual cyclicality can yield high apparent catamenial epilepsy rates, especially with short windows and ratio-only rules. Diagnostic studies should prespecify analyzability, comparator handling, and reproducibility requirements.

Classification of Evidence: Not applicable; this is a simulation study and does not evaluate a therapeutic or diagnostic intervention.

41 Search Terms: catamenial epilepsy; epilepsy; seizures; menstrual cycle; simulation study.

42 Neurology taxonomy numbers should be selected in the submission system.

43

44 **Introduction**

45 Catamenial epilepsy describes seizure exacerbation in relation to the menstrual cycle. The
46 clinical literature has long recognized perimenstrual, periovulatory, and inadequate luteal-phase
47 patterns, but reported prevalence varies widely because studies use different phase definitions,
48 thresholds, diary durations, and eligibility rules. Herzog and colleagues reported 39.1%
49 catamenial epilepsy in women who charted seizures and menses for three cycles using a two-of-
50 three-cycle rule, and later work from the NIH Progesterone Treatment Trial has been interpreted
51 as supporting prevalence near 44% in selected trial participants.^{1,5}
52 The diagnostic problem is not only biological. Seizures are themselves temporally structured:
53 clustering, circadian rhythms, and multidien rhythms have been documented in long-term diaries
54 and intracranial recordings.⁹⁻¹¹ If seizure cycles are realistic but independent of menstrual
55 cycles, finite diary windows can still contain chance alignments. Ratio definitions are especially
56 vulnerable when seizure counts are sparse or comparator phases contain few events. This creates
57 a null question that has direct implications for clinical research: what apparent catamenial
58 epilepsy rate should be expected even when no menstrual coupling exists?
59 We conducted a reproducible null-simulation study to estimate false-positive catamenial epilepsy
60 classification across definitions and observation windows. The analysis was restricted to Paper 1
61 questions: independent realistic seizure and hormone-cycle simulations, no true coupling, no
62 wavelet or hidden Markov modeling, and no treatment-effect simulation.

63 **Methods**

64 Study design. This was a prespecified Monte Carlo null simulation. For each synthetic
65 participant, seizure and menstrual/hormone diaries were generated independently for 36 months.

66 A participant-specific random circular day shift was applied before merging to prevent hidden
67 start-date alignment. All analyses were stratified by cohort; cohorts were never pooled.

68 Cohorts. The healthy ovulatory cohort included 50,000 synthetic participants intended to mimic a
69 cleaner research environment, with age 18-45 years and ovulatory cycling enforced through the
70 simulator adapter where available. The broad population cohort included 50,000 synthetic
71 participants across a broader menstruating-age range, age 13.0-54.9 years, allowing anovulation
72 and irregularity. Because the hormone-cycle simulator exposed medical-factor knobs but not a
73 natural prevalence sampler, PCOS, peri-menarche, perimenopause, and dysmenorrhea were
74 sampled from configuration rates.

75 Diary simulators. CHOCOLATES generated realistic non-catamenial seizure diaries using its
76 default seizure cyclicity and burden behavior. HORMONE-CYCLE generated independent
77 menstrual/hormone diaries. Thin adapters inspected available simulator APIs and retained useful
78 latent or observed participant summaries. The master seed was 20260505. Processing was
79 chunked and parallelized, and daily audit rows were saved for a 1% random sample.

80 Phase labeling and definitions. Herzog phases were assigned on the full diary before subsetting.
81 In a complete cycle of length L , day 1 was menstrual-flow onset and backward day was $d-(L+1)$.
82 Menstrual phase comprised days 1-3 and backward days -3 to -1; follicular phase days 4-9;
83 ovulatory phase days ≥ 10 and backward days ≤ -13 ; luteal phase backward days -12 to -4.

84 Exact Herzog 2004 was restricted to complete 3-cycle windows and required at least two of three
85 cycles to show any positive pattern. Windowed Herzog thresholds pooled phase-specific average
86 daily seizure frequencies across arbitrary windows. Minimum-data rules required at least 4
87 calendar months or 6 complete cycles and at least 4 seizure days. Reproducibility rules required
88 at least two-thirds of eligible cycles to be positive and a corresponding pooled ratio. Regression

used within-participant negative-binomial models, with Poisson robust fallback recorded when needed. Historical definitions were flagged as assumption-based operationalizations. Statistical analysis. The primary outcome was the person-window false-positive rate for any catamenial epilepsy under the null. Wilson 95% confidence intervals were calculated for simple proportions. Undefined ratios were never coerced to negative: positive numerator with zero comparator was infinite and positive; zero numerator with zero comparator was indeterminate. Study-level Monte Carlo simulated 10,000 studies of 30 participants per cohort using random 3-month windows and estimated probabilities that apparent prevalence exceeded 39.1% and 44.2%. Standard protocol approvals, registrations, and patient consents. This study used only synthetic participants and did not use human-subject data, identifiable patient information, images, or videos. Institutional review board approval and patient consent were not required for this simulation study.

Results

Simulation cohort and diary characteristics. The full run included 100,000 synthetic participants and 900,000 prespecified participant-window rows. Cohort summaries are shown in Table 1. The healthy ovulatory cohort had 100.0% ovulatory cycles by design. The broad population cohort had 79.0% ovulatory cycles, greater cycle-length variability, and similar observed seizure burden.

Primary full-diary false-positive rates. In 36-month windows, the windowed Herzog threshold definition classified 11.2% of healthy ovulatory participants and 36.3% of broad population participants as false-positive catamenial epilepsy (Table 2). Adding minimum-data rules changed the full-window estimates modestly because most full diaries were analyzable. Negative-

binomial regression reduced false-positive rates to approximately 4% in both cohorts. In the broad population cohort, the high windowed false-positive rate was driven largely by C3 logic: full-window C3 positivity occurred in 37.0% of applicable broad population windows, compared with C1 and C2 rates of 8.4% and 5.6%.

Observation-window effects. Short diary windows were the most vulnerable to apparent catamenial classification. In random 3-month calendar windows, windowed Herzog thresholds yielded false-positive rates of 41.3% in the healthy ovulatory cohort and 51.2% in the broad population cohort. Minimum-data, reproducibility, and regression definitions were intentionally indeterminate for 3-month calendar windows because the required minimum structure was absent. Exact Herzog 2004 applied to 3 complete cycles yielded false-positive rates of 50.3% and 51.6% among classifiable windows, while 53.8% and 66.7% of windows were indeterminate. Study-level Monte Carlo. In 10,000 simulated studies of 30 participants using 3-month windows and the windowed Herzog threshold rule, the mean apparent prevalence among all 30 participants was 37.5% in the healthy ovulatory cohort and 46.0% in the broad population cohort. Apparent prevalence was at least 39.1% in 45.9% and 79.9% of null studies, respectively, and at least 44.2% in 19.7% and 54.5%.

Null cycle-day profile. The daily audit sample did not show a coherent menstrual-cycle seizure profile consistent with true coupling (Figure 3), supporting that the observed classifications arose from finite-window chance alignment and rule behavior rather than simulated hormonal effects.

Discussion

This null simulation shows that apparent catamenial epilepsy can arise frequently even when seizure and menstrual processes are generated independently. The magnitude depends strongly on the definition, window length, analyzability rules, and cohort menstrual-cycle assumptions.

Ratio-only windowed definitions were vulnerable, particularly in short windows and in the broad population cohort where anovulatory-cycle logic made C3 classification common. Regression and reproducibility requirements reduced false positives but did so partly by increasing indeterminate windows or requiring longer diaries.

These results help interpret the broad range of prevalence estimates in the catamenial epilepsy literature. Earlier work established biologically plausible and clinically important menstrual-cycle seizure patterns, including C1, C2, and C3 patterns.¹⁻⁵ Treatment literature also suggests that robust perimenstrual exacerbation may identify subgroups of interest, even though randomized evidence for hormonal therapy remains limited and mixed.^{6,7} Our results do not challenge the existence of true catamenial epilepsy. Instead, they quantify a diagnostic background rate expected from chance alignment when realistic seizure cycles and realistic menstrual cycles coexist.

The findings are also consistent with modern seizure-cycle research. Multidien seizure rhythms and seizure forecasting studies have shown that seizure risk varies over days to weeks, and those rhythms can be detected in long-term electronic diaries and chronic recordings.⁹⁻¹¹ A menstrual-cycle analysis that does not account for intrinsic seizure cyclicity can therefore mistake phase overlap for hormone coupling. This is most problematic when the observation window is short, because a few events can dominate average daily seizure-frequency ratios.

Several practical implications follow. First, catamenial epilepsy studies should report both positive rates and indeterminate rates, because stricter definitions may appear more conservative partly by declining to classify insufficient windows. Second, comparator phases with zero seizures require explicit rules; silently treating undefined ratios as negative can bias estimates.

Third, short windows should be interpreted cautiously unless the diagnostic objective is

screening rather than classification. Fourth, C3 classification in heterogeneous or irregular-cycle populations requires particularly careful hormone or ovulatory-status ascertainment.

This study has limitations. The results are only as realistic as the CHOCOLATES and HORMONE-CYCLE simulators and the adapter assumptions. The broad population medical-factor rates were configuration-sampled rather than drawn from a simulator-native prevalence sampler. The analysis did not simulate true menstrual coupling, treatment effects, wavelet analyses, or hidden Markov models. We also did not adjudicate clinical seizure types or medication changes. These restrictions were deliberate for a Paper 1 null study and should be addressed in later work that introduces calibrated true-coupling scenarios.

In conclusion, under a realistic null model, false-positive catamenial epilepsy classification can be common, sometimes reaching benchmark prevalence values reported in clinical literature. Future diagnostic and interventional studies should prespecify diary length, menstrual-phase labeling, analyzability thresholds, ovulatory-status handling, and reproducibility criteria before interpreting apparent menstrual clustering as evidence of hormone-linked seizure exacerbation.

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Disclosure

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181 Data Availability

182 The simulation code, configuration, analysis outputs, and machine-readable manifest are
183 available in the project repository. The manuscript should be updated with a public archive DOI
184 or repository URL before submission.

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212 Tables

213 Table 1. Simulated cohort characteristics.

Cohort	N	Mean age, y	Mean cycle length, d	Cycle-length SD, d	Ovulatory cycles	Seizure days/month	Seizures/month
healthy ovulatory	50,000	31.5 (SD 7.8)	29.15	3.39	100.0%	2.46	6.83
broad population	50,000	34.0 (SD 12.1)	30.92	4.74	79.0%	2.45	6.80

214

215 Values are cohort-level means from the full 100,000-participant run. Cycle-length SD is the

216 mean within-participant cycle-length SD.

217 Table 2. Key null false-positive results.

Cohort	Analysis setting	Definition	Classifiable denominator	Main result	Indeterminate	Benchmark probabilities
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.2% (11.0%-11.5%)	0.8%	Not applicable
healthy ovulatory	36-month full diary	Windowed Herzog with minimum data	48,359 / 50,000	10.2% (9.9%-10.5%)	3.3%	Not applicable
healthy ovulatory	36-month full diary	Negative-binomial regression	48,359 / 50,000	4.1% (3.9%-4.2%)	3.3%	Not applicable
healthy ovulatory	3 complete cycles	Exact Herzog 2004, any pattern	23,101 / 50,000	50.3% (49.6%-50.9%)	53.8%	Not applicable
healthy ovulatory	10,000 n=30 studies; 3-month windows	Windowed Herzog thresholds	All 30 participants	Mean apparent prevalence 37.5%	Not applicable	45.9%; 19.7%
broad population	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	36.3% (35.8%-36.7%)	0.8%	Not applicable
broad population	36-month full diary	Windowed Herzog with minimum data	48,333 / 50,000	35.6% (35.2%-36.0%)	3.3%	Not applicable
broad population	36-month full diary	Negative-binomial regression	48,333 / 50,000	4.2% (4.0%-4.4%)	3.3%	Not applicable
broad population	3 complete cycles	Exact Herzog 2004, any pattern	16,673 / 50,000	51.6% (50.8%-52.4%)	66.7%	Not applicable

Cohort	Analysis setting	Definition	Classifiable denominator	Main result	Indeterminate	Benchmark probabilities
broad population	10,000 n=30 studies; 3-month windows	Windowed Herzog thresholds	All 30 participants	Mean apparent prevalence 46.0%	Not applicable	79.9%; 54.5%

218

219 Main result is false-positive rate for participant windows unless the row is labeled as a study-

220 level Monte Carlo result. Benchmark probabilities are P(apparent prevalence at least 39.1%) and

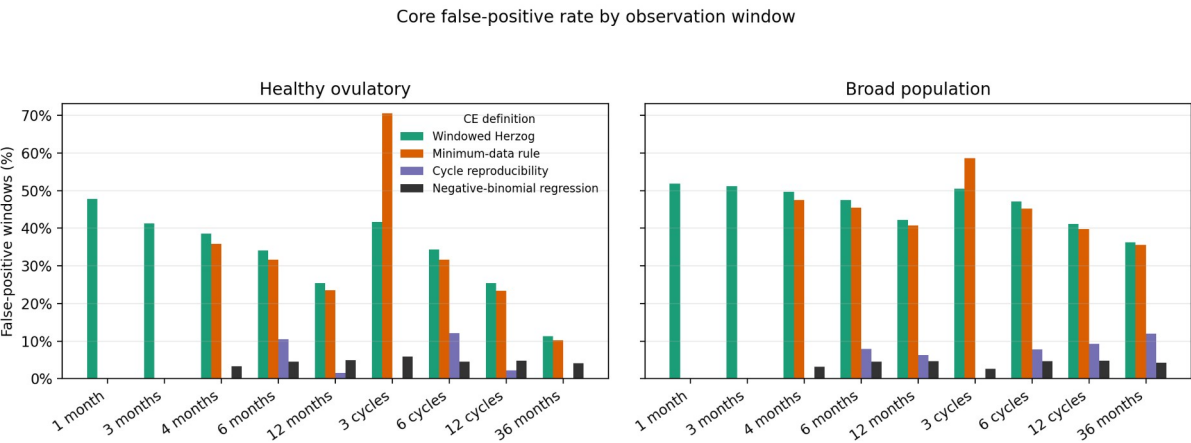
221 P(apparent prevalence at least 44.2%).

222

223 **Figure Legends**

224 Figure 1. False-positive rates by observation window.

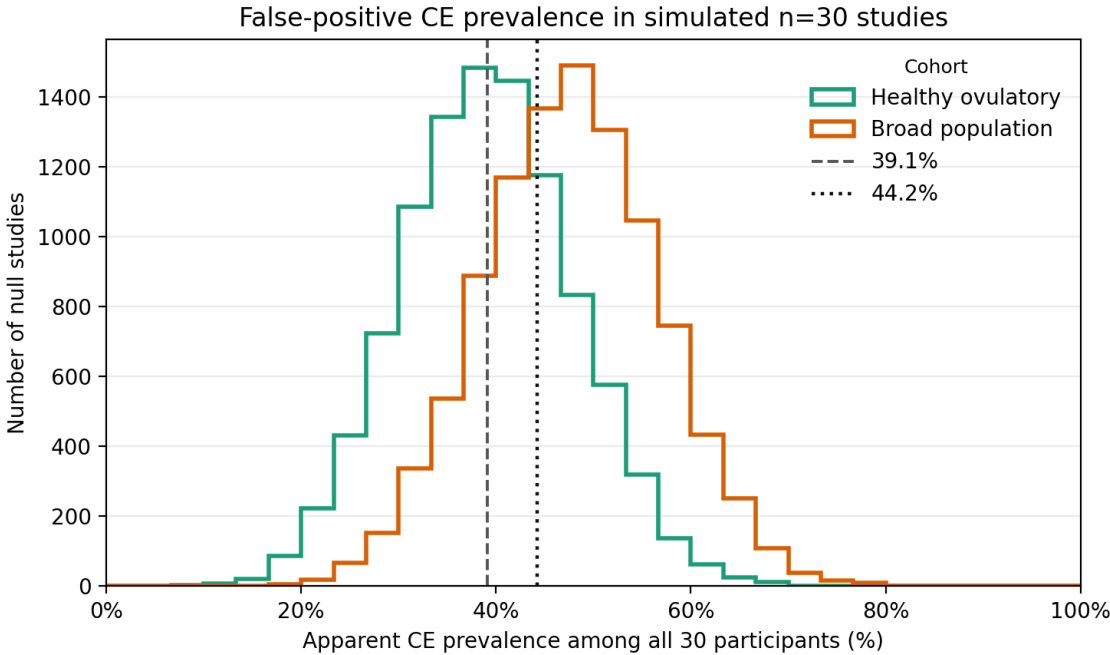
225 Bars show the percentage of classifiable participant windows classified as catamenial epilepsy
226 under the null. Panels are separated by cohort; definitions are shown in the legend.



227

228 Figure 2. Apparent prevalence in null studies.

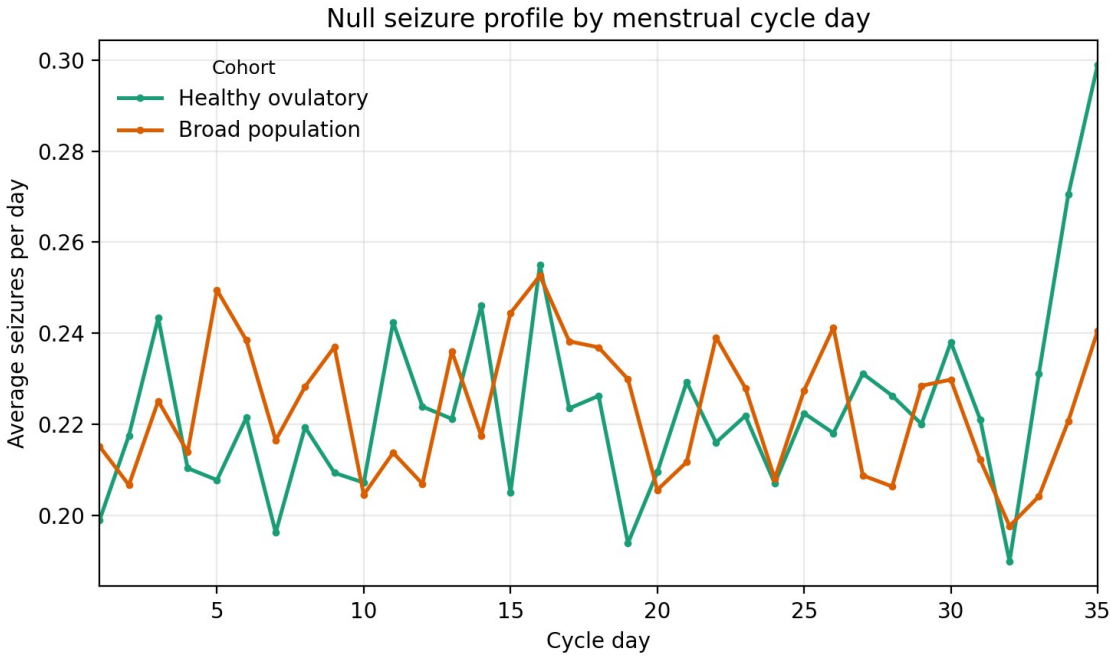
229 Distribution of apparent catamenial epilepsy prevalence in 10,000 simulated studies of 30
230 participants using random 3-month windows and the windowed Herzog threshold definition.
231 Vertical lines show benchmark prevalence values of 39.1% and 44.2%.



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233 Figure 3. Null seizure profile by cycle day.

234 Average daily seizure frequency by menstrual cycle day in the 1% daily audit sample. The
235 independent null simulation should not produce a coherent hormone-coupled seizure profile.



236